

Worksheet

1. Does the sequence $\left\{ \left(\frac{4}{5} \right)^k \right\}$ converge? If so what is the limit? If not, why not?
2. Does the series $\sum_{k=1}^{\infty} \left(\frac{4}{5} \right)^k$ converge? If so what does it converge to? If not, why not?
3. Does $\sum_{k=3}^{\infty} \ln \left(\frac{n}{n+1} \right)$ converge? Why or why not?
4. Does $\sum_{k=1}^{\infty} \frac{2^{k-2}}{7^{k+1}}$ converge? If so what does it converge to? If not, why not?